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Linux Administration
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sed

SED is a command-line text-processing tool that allows us to search, substitute, delete, and modify text in files. It processes text line by line and uses commands and regular expressions to select and manipulate specific lines or patterns.

Exercise 1

Task

- Change the name Jo to Josephine

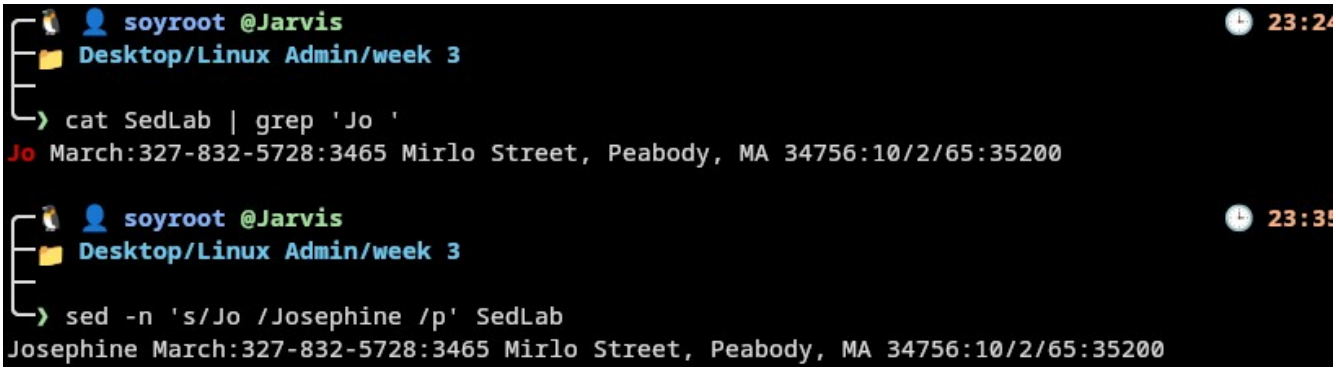
Command

- sed -n 's/Jo /Josephine /p' SedLab

How it works

- The sed command is used to search, substitute, and modify text in files. In this command, I use the -n option to prevent sed from printing all the lines by default. Inside single quotes, I specify the substitution expression as one argument. The s indicates that I want to perform a substitution, Jo is the text I am searching for, and Josephine is the new text that will replace it. The spaces after both names are included in the expression to match Jo as a separate name. The p prints the line where the substitution was made, and SedLab specifies the file being processed.

Result



Exercise 2

Task

- Delete the last 5 lines.

Command

- sed -e :a -e '$d;N;2,5ba' -e 'P;D' SedLab

How it works

- This command uses sed to remove the last five lines without needing to know how many lines the file has. First, we use -e to specify multiple expressions. Then we use the : command to create a label named a, which allows us to return to that point using the b command. The $ symbol represents the last line of the file, and d tells sed to delete the current pattern space when it reaches that line. We use N to read the next line and add it to the pattern space, where sed temporarily keeps the lines. The 2,5ba tells sed to return to our label while processing lines 2 through 5, allowing us to accumulate lines before printing. After that, P prints the first line from the pattern space, and D removes that first line while keeping the remaining lines. This process continues until sed reaches the end of the file, leaving the last five lines out of the output. Finally, SedLab specifies the file we are working with.

Result

```
soyroot @Friday Desktop/Linux Admin/week 3 master[?] 00:56
❯ sed -e :a -e '$d;N;2,5ba' -e 'P;D' SedLab
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300

Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
```

Exercise 3

Task

- Print lines 3 through 15.

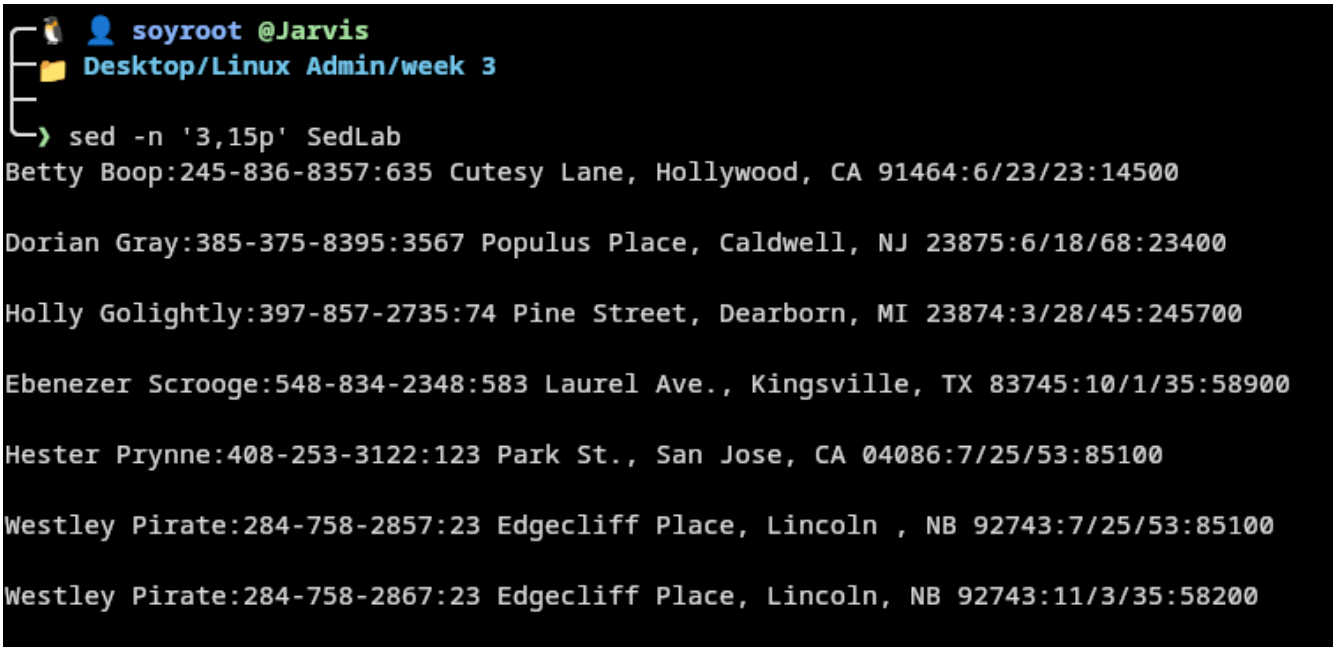
Command

- sed -n '3,15p' SedLab

How it works

- This command uses sed to print only lines 3 through 15 from the file. First, we use the -n option to prevent sed from printing every line automatically. Then we specify the range 3,15 to tell sed where to start and stop. The p command tells sed to print the lines within that range. Finally, SedLab is the file we are working with. This allows us to display only the lines we need without changing the original file.

Result



Exercise 4

Task

- Delete lines for people who live in California

Command

- sed '/CA/d' SedLab

How it works

- This command uses sed to delete the lines for people who live in California. First, we use /CA/ to search for the lines containing CA, which represents California in our file. The forward slashes are used to specify the pattern we want to search for. Then we use d to delete the lines that match that pattern. By default, sed prints the rest of the lines that do not match, so we don't need to use p. Finally, SedLab specifies the file we are working with.

Result



Exercise 5

Task

- Print all lines where the birthdays are in the first week of the month. Be careful of the dates for birthdays, the format is MM/DD/YY

Command

- sed -n '//[1-7]//p' SedLab

How it works

- This command uses sed to print the lines where the birthdays fall within the first week of the month. First, we use -n to prevent sed from printing all the lines automatically. Then we use the forward slashes to define the pattern we want to search for. Inside the pattern, we use / to represent a literal forward slash because the dates in our file use / as a separator. The brackets [1-7] specify a range of numbers from 1 through 7, which represent the first seven days of the month. We add another / after the range to identify the end of the day portion of the date. Finally, p tells sed to print the lines that match our pattern, and SedLab specifies the file we are working with.

Result



Exercise 6

Task

- Append three asterisks (*) to the end of lines starting with Sir

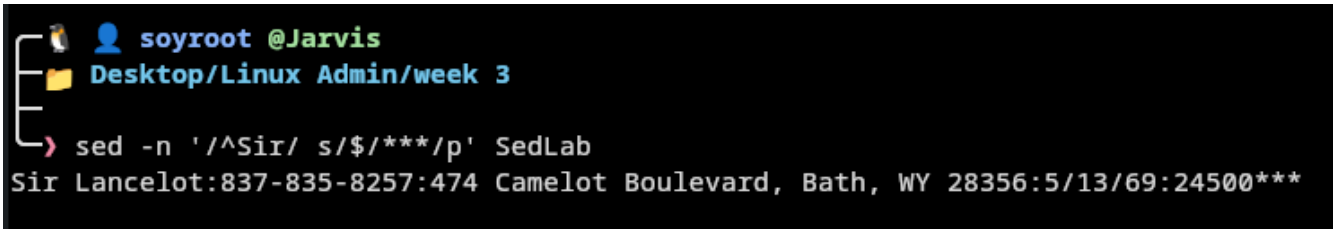
Command

- sed -n '/^Sir/ s/$/***/p' SedLab

How it works

- This command uses sed to add three asterisks at the end of the lines that start with Sir. First, we use -n to prevent sed from printing every line automatically. Then we use /^Sir/ to search for the lines that begin with Sir. The ^ symbol represents the beginning of a line, and the forward slashes define the pattern we are searching for. After that, we use s to perform a substitution. Inside the substitution, $ represents the end of the line, and *** is the text we want to add at that position. Finally, p tells sed to print the lines where the substitution was made, and SedLab specifies the file we are working with. The original file remains unchanged.

Result



Exercise 7

Task

- Replace the line containing ”Westley Pirate” with the phrase ”As you wish.” Make sure you replace the whole line not just the name

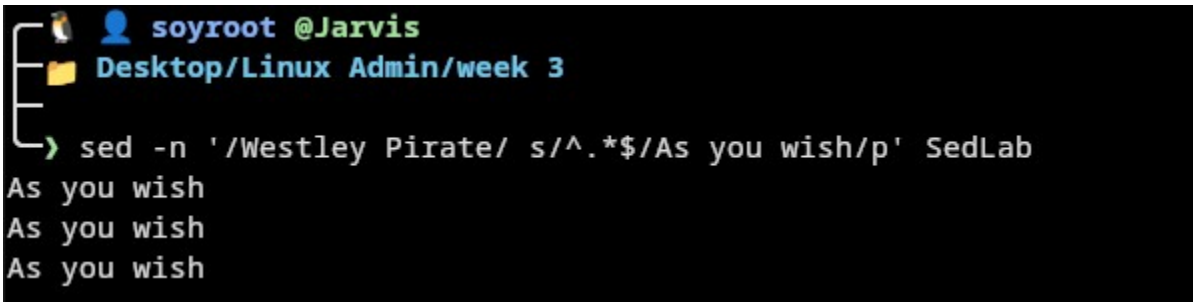
Command

- sed -n '/Westley Pirate/ s/^.*$/As you wish/p' SedLab

How it works

- This command uses sed to replace the entire line containing Westley Pirate with the phrase As you wish. First, we use -n to prevent sed from printing every line automatically. Then we use /Westley Pirate/ to search for the line containing that name. The forward slashes define the pattern we are looking for. After that, we use s to perform a substitution. The ^ symbol represents the beginning of the line, the dot . matches any character, and the * means zero or more of the previous character. Finally, $ represents the end of the line. By combining ^.*$ we can match the entire line instead of replacing only the name. Then we specify As you wish as the replacement text and use p to print the result. SedLab is the file we are working with.

Result



Exercise 8

Task

- Change Minerva McGonagall’s birthday to 10/04/1935. Assume you don’t know Minerva’s original birthday. Use a regular expression to search for it.

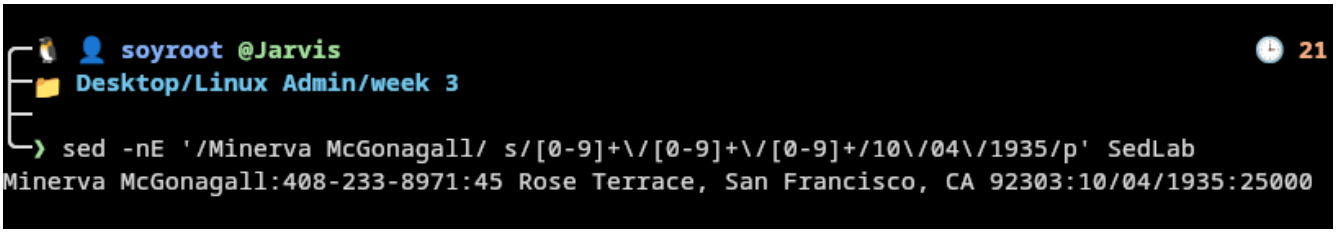
Command

- sed -nE '/Minerva McGonagall/ s/[0-9]+/[0-9]+/[0-9]+/10/04/1935/p' SedLab

How it works

- This command uses sed to change Minerva McGonagall's birthday without needing to know her original date. First, we use -n to prevent sed from printing all the lines automatically, and -E allows us to use extended regular expressions. Then we use /Minerva McGonagall/ to find the line containing her name. After that, we use s to perform a substitution. Inside the regular expression, [0-9] represents any digit from 0 through 9, and the + means one or more digits. We use this pattern three times to match the month, day, and year. The / represents the forward slashes separating each part of the date. Once sed finds the date, we replace it with 10/04/1935. Finally, p prints the modified line, and SedLab specifies the file we are working with.

Result



Exercise 9

Task

- Delete all blank lines.

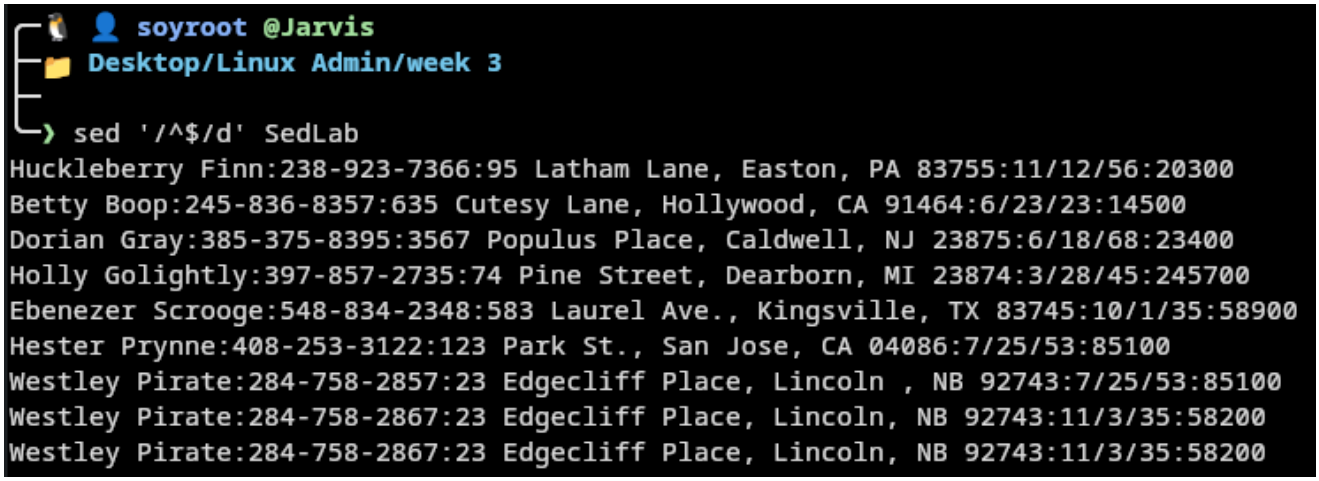
Command

- sed '/^$/d' SedLab

How it works

- This command uses sed to delete all the blank lines from the output. First, we use the forward slashes to define the pattern we want to search for. Inside the pattern, ^ represents the beginning of the line, and $ represents the end of the line. Since there is nothing between these two symbols, sed identifies lines that are completely empty. Then we use d to delete the lines that match this pattern. By default, sed prints the remaining lines, so we don't need to use p. Finally, SedLab specifies the file we are working with.

Result



Exercise 10

Task

- Write a sed script that will (actual sed script, NOT just the commands on the command line)

  (a) Insert above the first line the title - Great Literary Characters -.

  (**b)** Print the contents of the file, but instead of the phone number starting with an area code, have it start with a 1+, then include the area code and number. For example, (603)123-1234 would turn into 1+(603)123-1234

  (c) Append at the end of the file ”Happily Ever after. The End”

Command

- nano scriptsed.sed

Sed Script

```
1i\ - Great Literary Characters       -

s/:/:1+/

$a\Happily Ever after.The End
```

Command to Run the Script

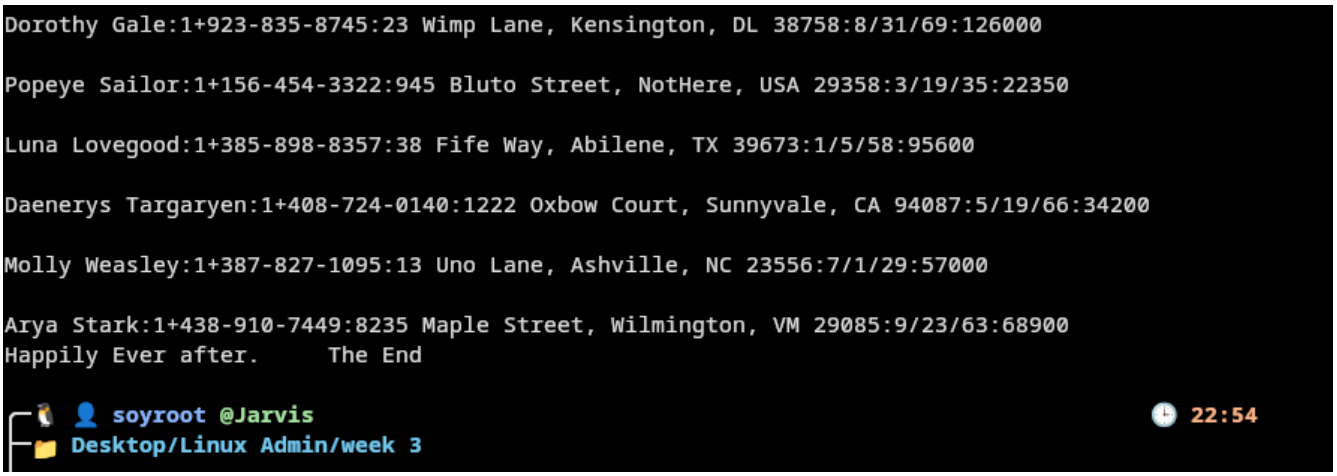
- sed -f scriptsed.sed SedLab

How it works

- For this exercise, we need to create a sed script instead of writing all the commands directly in the terminal. First, we create a file named scriptsed.sed using nano, which allows us to write and save our sed instructions. The first instruction uses 1 to specify the first line of the file and i\ to insert the title Great Literary Characters before that line. Next, we use s/:/:1+/ to add 1+ before the phone number. In our file, the colon separates the name from the phone number, so we replace the first colon with :1+. By default, sed replaces only the first matching occurrence on each line. For the last instruction, we use $ to identify the last line and a\ to append Happily Ever after.The End after it. Once the script is saved, we run it using sed -f scriptsed.sed SedLab. The -f option tells sed to read the instructions from our script file, and SedLab specifies the file we want to process. This allows us to perform all three operations using one script without modifying the original file.

Result


soyroot @Jarvis
22:54
Desktop/Linux Admin/week 3
sed -f scriptsed.sed SedLab
- Great Literary Characters -
Huckleberry Finn:1+238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:1+245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:1+385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400
Holly Golightly:1+397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:1+548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900


Dorothy Gale:1+923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:1+156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:1+385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:1+408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:1+387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:1+438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
Happily Ever after. The End
soyroot @Jarvis
22:54
Desktop/Linux Admin/week 3

Sources

Holden-Gouveia, Adrianna. Sed Linux Administration.

https://www.aholdengouveia.name/LinuxAdmin/Sed.html

GNU Project. GNU sed Manual. Free Software Foundation.

https://www.gnu.org/software/sed/manual/sed.html

Goyvaerts, Jan. Regular-Expressions.info: Regex Tutorial.

https://www.regular-expressions.info/tutorial.html